Week 5 Lab: Sensors & ADC Applications

EE0827 Embedded System Applications

Overview

In this lab, you will interface two analog sensors with the STM32 ADC: a K-type thermocouple with an LM741 amplifier circuit and an LDR (Light Dependent Resistor). You will null the amplifier offset, implement digital filtering, and optionally perform 2-point calibration to convert ADC readings to temperature.

Time allocation: 90 minutes

- Part 1 - CubeMX Configuration: 15 min
- Part 2 - Thermocouple + Amplifier: 30 min
- Part 3 - Light Sensor: 20 min
- Part 4 - Digital Filtering: 15 min
- Part 5 - Two-Point Calibration *(optional)*: 10 min

Learning Objectives

By the end of this lab, you will be able to:

1. Configure the STM32 ADC for multi-channel operation
2. Interface a thermocouple sensor with external amplification to the ADC
3. Design and implement voltage dividers for resistive sensors (LDR)
4. Implement a moving average filter in firmware
5. *(Optional)* Perform 2-point calibration and apply calibration coefficients

Safety Gate

Safety Checklist (Complete Before Powering On)

- ☐ Bench supply set to ±12V (series mode) before powering on the LM741
- ☐ Thermocouple center wire to pin 3 (+), brass body to GND
- ☐ No shorts between breadboard rows
- ☐ USB cable connected for programming and UART output

Sensor Handling

1. **Verify amplifier output with multimeter** before connecting to STM32 PA0 — must be below 3.3V
2. **Keep analog wires short** — thermocouple signals are in the µV range and pick up noise easily
3. **Do not touch the thermocouple tip** during measurements — body heat will shift the reading

Required Parts and Tools

Hardware

- STM32 Nucleo-F401RE board
- Breadboard and jumper wires (~10)
- 10kΩ resistor (for LDR voltage divider)
- K-type thermocouple probe (stove/oven type)
- LM741 op-amp (DIP-8)
- 220kΩ resistor (R_f), 1kΩ resistors ×2 (R_g and R_p)
- 5–10kΩ potentiometer (offset null)
- Bench power supply (±12V, series mode)
- LDR photoresistor (1kΩ-100kΩ range)
- USB cable (programming and UART)

Tools

- Multimeter (voltage verification)
- Oscilloscope (noise measurement)
- Serial terminal (PuTTY, TeraTerm, or CubeIDE terminal)

! PRE-LAB SECTION

Complete the following BEFORE arriving at the lab session.

Pre-Lab Checklist

Complete before arriving at the lab:

- ☐ Read theory slides on ADC operation and sensor interfacing
- ☐ Understand how a thermocouple generates voltage (~41µV/°C for K-type)
- ☐ Understand why external amplification is needed for thermocouple signals
- ☐ Understand voltage divider calculations
- ☐ Complete all pre-lab calculations below

Pre-Lab Questions

1. A K-type thermocouple outputs ~41µV/°C. What is the ADC's voltage resolution at 12-bit / 3.3V? Why can't the STM32 read the thermocouple directly?

_____

2. Why do resistive sensors (like LDR) need a voltage divider while voltage-output sensors (like the amplifier) don't? _____

3. What is the purpose of a moving average filter in sensor applications? _____

Pre-Lab Calculations

1. Thermocouple + Amplifier Output

A K-type thermocouple outputs approximately 41µV per degree Celsius. The LM741 non-inverting amplifier has gain $A_v = 1 + R_f/R_g = 1 + 220\text{k}/1\text{k} = 221$.

At room temperature (25°C), what is the raw thermocouple voltage?

$$V_{TC} = 25°C \times 41\,\mu V/°C = \_\_\_\_\_ mV$$

The STM32 ADC resolution is $3.3V/4095 \approx 0.806\,mV$. What minimum amplifier gain is needed to produce at least 1 LSB per °C?

$$A_{v,min} = \frac{0.806\,mV}{41\,\mu V} = \_\_\_\_\_$$

With $A_v = 221$, what amplifier output voltage do you expect at 25°C?

$$V_{out} = V_{TC} \times A_v = \_\_\_\_\_ mV$$

What ADC value would that correspond to? (V_{REF} = 3.3V, 12-bit)

$$ADC = \frac{V_{out}}{3.3V} \times 4095 = \_\_\_\_\_$$

2. LDR Voltage Divider

Using a 10kΩ resistor as R_{fixed} (top of divider) with LDR on bottom:

$$V_{out} = V_{cc} \times \frac{R_{LDR}}{R_{fixed} + R_{LDR}}$$

Calculate V_{out} for the following LDR resistance values:

Light Condition	R_{LDR}	V_{out} (calculated)
Bright	1kΩ	_____ V
Room light	10kΩ	_____ V
Dark	100kΩ	_____ V

3. Filter Design

For 10 Hz sampling with 8-sample moving average, how many seconds for full response?

$$t_{response} = \frac{8 samples}{10 Hz} = \_\_\_\_\_ s$$

IN-LAB SECTION

The following sections are completed during the lab session.

Part 1

CubeMX ADC Configuration *15 minutes*

Step 1.1: Create New Project

1. Open STM32CubeIDE
2. Create new STM32 project for Nucleo-F401RE
3. Name it `W05_Sensors_ADC`

Step 1.2: Configure ADC Channels

In the Pinout view:

1. Click on **PA0** → Select **ADC1_IN0** (Temperature sensor)
2. Click on **PA1** → Select **ADC1_IN1** (Light sensor)

Step 1.3: Configure ADC Parameters

Navigate to **Analog** → **ADC1** in the left panel:

Parameter	Value
Clock Prescaler	PCLK2 divided by 4
Resolution	12 bits
Data Alignment	Right
Scan Conversion Mode	Enabled
Number of Conversion	2

Rank 1: Channel 0, Sampling Time 84 Cycles

Rank 2: Channel 1, Sampling Time 84 Cycles

Step 1.4: Configure UART

Navigate to **Connectivity** → **USART2**:

Parameter	Value
Mode	Asynchronous
Baud Rate	115200

Step 1.5: Generate Code

1. Save the IOC file
2. Click **Project → Generate Code**

Checkpoint 1

Verify your CubeMX configuration before proceeding.

Part 2

Thermocouple + Amplifier Interface *25 minutes*

Step 2.1: Hardware Setup

Wire the K-type thermocouple through the LM741 non-inverting amplifier to the STM32:

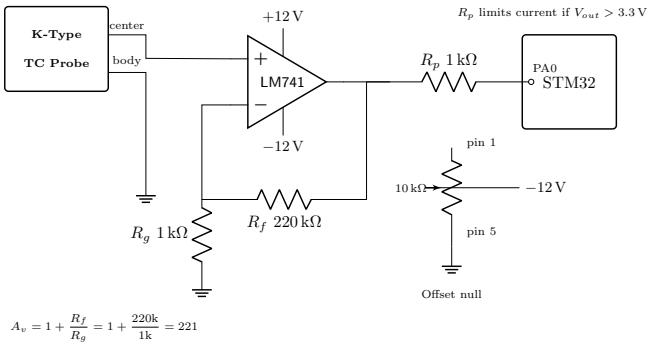

Figure 1: LM741 thermocouple amplifier circuit

The stove thermocouple is a coaxial probe — the two dissimilar metals share a single cable. The **center conductor** is one lead and the **brass body/housing** is the other.

Connection	Details
TC center wire	LM741 pin 3 (non-inverting input, +)
TC brass body	Circuit GND
R_f (220kΩ)	Pin 6 (output) to pin 2 (inverting input, −)
R_g (1kΩ)	Pin 2 to GND
Pin 7 (V+)	+12V from bench supply
Pin 4 (V−)	−12V from bench supply
R_p (1kΩ)	Pin 6 to PA0 (output protection)
Offset null pot (5–10kΩ)	Between pins 1 and 5, wiper to −12V

Verify Before Connecting to MCU

Power the amplifier and measure pin 6 (output) with a multimeter **before** connecting R_p to PA0. The output must be below 3.3 V at room temperature. If not, check wiring or reduce gain.

Step 2.2: Null Offset and Verify

1. Set bench supply to ±12V (series mode, both channels at 12V)
2. **Short the thermocouple input** — connect center wire to brass body (forces 0mV)
3. Measure LM741 output (pin 6) with a multimeter
4. Turn the offset null pot until output reads as close to **0.000V** as possible
5. Remove the short, reconnect the thermocouple
6. Warm the thermocouple tip with your fingers — output should increase
7. **Once confirmed < 3.3V**, connect R_p output to PA0

Measurement	Value
Output after offset null (shorted input)	_____ mV
Output at room temp	_____ mV
Output with finger heat	_____ mV

Why offset null matters

The LM741 has an inherent input offset voltage of 1–6 mV. At a gain of ~220, this becomes 220–1320 mV at the output — comparable to the thermocouple signal itself. The offset null pot cancels this error *before* amplification.

Step 2.3: Basic ADC Reading Code

Add to `main.c` (in USER CODE sections):

```
/* USER CODE BEGIN Includes */
#include <stdio.h>
#include <string.h>
/* USER CODE END Includes */

/* USER CODE BEGIN PV */
uint16_t adc_temp_raw = 0;
float temp_voltage = 0;
char uart_buf[64];
/* USER CODE END PV */

/* USER CODE BEGIN 0 */
uint16_t read_adc_channel(uint8_t channel) {
    ADC_ChannelConfTypeDef sConfig = {0};
    sConfig.Channel = (channel == 0) ? ADC_CHANNEL_0 : ADC_CHANNEL_1;
```

```
    sConfig.Rank = 1;
    sConfig.SamplingTime = ADC_SAMPLETIME_84CYCLES;

    HAL_ADC_ConfigChannel(&hadc1, &sConfig);
    HAL_ADC_Start(&hadc1);
    HAL_ADC_PollForConversion(&hadc1, 10);
    return HAL_ADC_GetValue(&hadc1);
}
/* USER CODE END 0 */
```

Step 2.4: Main Loop Code

Although the gain is known ($A_v \approx 221$), the cold junction offset and component tolerances make a direct voltage-to-temperature formula unreliable. We display raw ADC and voltage here; calibrated temperature conversion is covered in Part 5.

```
while (1) {
    adc_temp_raw = read_adc_channel(0);
    temp_voltage = (adc_temp_raw / 4095.0f) * 3.3f;

    sprintf(uart_buf, "TC: ADC=%4d, V=%.3fV\r\n",
            adc_temp_raw, temp_voltage);
    HAL_UART_Transmit(&huart2, (uint8_t *)uart_buf,
                      strlen(uart_buf), HAL_MAX_DELAY);
    HAL_Delay(500);
}
```

Step 2.5: Test Thermocouple Response

1. Build and flash the project
2. Open serial terminal (115200 baud)
3. Pinch the thermocouple tip — ADC value should increase
4. Release and blow on it — ADC value should decrease

Measurement	Value
ADC raw at room temp	_____
ADC raw with finger heat	_____

Checkpoint 2

Verify that the ADC value changes when you heat/cool the thermocouple tip.

Part 3

Light Sensor Interface

20 minutes

Step 3.1: Hardware Setup

Build the LDR voltage divider circuit:

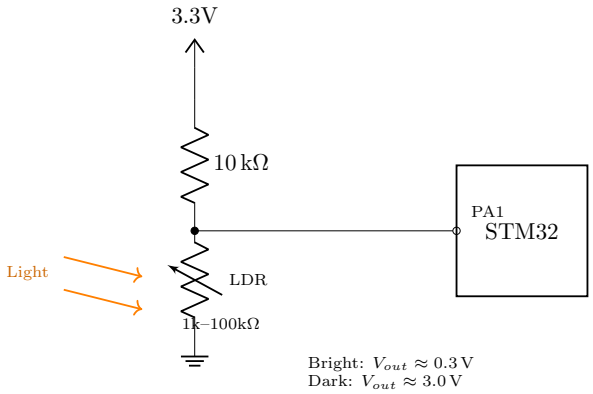

Figure 2: LDR voltage divider interface

Step 3.2: Verify Hardware

Measure voltage at PA1 junction with multimeter:

Condition	Expected	Measured
Bright light	~0.3V	_____ V
Room light	~1.65V	_____ V
Dark (covered)	~3.0V	_____ V

Step 3.3: Add Light Sensor Code

Add variables and update main loop:

```
/* USER CODE BEGIN PV */
uint16_t adc_light_raw = 0;
float light_voltage = 0;
float light_percent = 0;
/* USER CODE END PV */
```

```
while (1) {
    // Thermocouple (raw voltage - calibration comes in Part 4)
    adc_temp_raw = read_adc_channel(0);
    temp_voltage = (adc_temp_raw / 4095.0f) * 3.3f;

    // Light (invert: 0V=bright, 3.3V=dark)
    adc_light_raw = read_adc_channel(1);
    light_voltage = (adc_light_raw / 4095.0f) * 3.3f;
    light_percent = 100.0f - (light_voltage / 3.3f * 100.0f);

    sprintf(uart_buf, "TC: %.3fV | Light: %.0f%%\r\n",
            temp_voltage, light_percent);
    HAL_UART_Transmit(&huart2, (uint8_t *)uart_buf,
                      strlen(uart_buf), HAL_MAX_DELAY);
    HAL_Delay(500);
}
```

Checkpoint 3

Verify both sensors are working before proceeding.

Part 4

Digital Filtering

15 minutes

Step 4.1: Implement Moving Average Filter

```
/* USER CODE BEGIN PV */
#define FILTER_SIZE 8
uint16_t temp_buffer[FILTER_SIZE] = {0};
uint16_t light_buffer[FILTER_SIZE] = {0};
uint8_t filter_index = 0;
uint32_t temp_sum = 0, light_sum = 0;
/* USER CODE END PV */

uint16_t filter_update(uint16_t *buffer, uint32_t *sum, uint16_t sample) {
    *sum -= buffer[filter_index];
    buffer[filter_index] = sample;
    *sum += sample;
    return *sum / FILTER_SIZE;
}
```

Step 4.2: Update Main Loop with Filtering

```
while (1) {
    adc_temp_raw = read_adc_channel(0);
    uint16_t temp_filt = filter_update(temp_buffer, &temp_sum, adc_temp_raw);

    adc_light_raw = read_adc_channel(1);
    uint16_t light_filt = filter_update(light_buffer, &light_sum, adc_light_raw);

    filter_index = (filter_index + 1) % FILTER_SIZE;

    sprintf(uart_buf, "Raw: T=%4d L=%4d | Filt: T=%4d L=%4d\r\n",
            adc_temp_raw, adc_light_raw, temp_filt, light_filt);
    HAL_UART_Transmit(&huart2, (uint8_t *)uart_buf,
                      strlen(uart_buf), HAL_MAX_DELAY);
    HAL_Delay(100);
}
```

Step 4.3: Observe Filter Behavior

1. Touch LDR wire briefly — observe raw spikes, filtered stays smooth
2. Blow on thermocouple — observe gradual filtered response

Checkpoint 4

Verify filtering is working — raw values should jitter while filtered values stay smooth.

Part 5

Two-Point Calibration

10 minutes *(optional)*

To convert ADC counts into actual temperature, you need two known reference points. The calibration compensates for the amplifier gain, offset, and cold junction effects all at once.

Step 5.1: Collect Calibration Data

Record the filtered ADC value at two known temperatures:

Point	Method	Known Temp	Filtered ADC
1	Room temperature (read from thermometer or phone)	_____ °C	_____
2	Warm water or sustained finger contact	_____ °C	_____

Cold Junction Compensation

A thermocouple measures the *difference* between the hot junction (probe tip) and cold junction (where the wires meet the copper traces). Your 2-point calibration implicitly compensates for the cold junction offset at the current ambient temperature.

Step 5.2: Add Calibration to Code

```
/* USER CODE BEGIN PV */
// Fill in your measured calibration values:
float cal_adc_1 = 0;     // ADC at reference point 1 (room temp)
float cal_temp_1 = 0;    // Known temp at point 1 (°C)
float cal_adc_2 = 0;     // ADC at reference point 2 (warm)
float cal_temp_2 = 0;    // Known temp at point 2 (°C)
/* USER CODE END PV */

// In main loop, after filtering:
float temp_celsius = cal_temp_1
    + (temp_filt - cal_adc_1)
```

```
    * (cal_temp_2 - cal_temp_1)
    / (cal_adc_2 - cal_adc_1);

sprintf(uart_buf, "Temp: %.1f C (filt ADC=%d)\r\n",
        temp_celsius, temp_filt);
HAL_UART_Transmit(&huart2, (uint8_t *)uart_buf,
                  strlen(uart_buf), HAL_MAX_DELAY);
```

Step 5.3: Verify Calibration

1. Fill in your two calibration values and rebuild
2. Check that room temperature reads correctly on the serial terminal
3. Warm the probe — temperature should increase smoothly

Checkpoint 5 (optional)

Verify calibrated temperature output is reasonable at room temperature.

Measurement Summary

ID	Measurement	Expected	Measured	Pass
M1	Amplifier output after offset null (shorted)	~0mV		
M2	Amplifier output at room temp	100–500mV		
M3	ADC responds to finger heat	Increase visible		
M4	Filter smooths raw jitter	Filt < Raw variation		

Analysis Questions

1. Why does the LDR need a voltage divider but the thermocouple amplifier doesn't? _____

2. How would you modify the filter to respond faster? What's the trade-off? _

3. What happens to your calibration if the room temperature changes signifi-

cantly? (Hint: think about cold junction compensation.) _____

Completion Checklist

- ☐ Checkpoint 1: CubeMX configuration verified
- ☐ Checkpoint 2: Thermocouple + amplifier working
- ☐ Checkpoint 3: Both sensors working
- ☐ Checkpoint 4: Filtering demonstrated
- ☐ Checkpoint 5 *(optional)*: Calibrated temperature output verified
- ☐ All measurements recorded
- ☐ Analysis questions answered

Success Criteria

This lab is ungraded practice. Use these criteria to gauge your progress:

Area	What to Aim For
Pre-lab	All calculations completed before lab
ADC Configuration	CubeMX settings correct, code compiles
Thermocouple + Amplifier	Offset nulled, ADC responds to temperature changes
Light Sensor	Readings change with light level
Filtering	Smooth output, reduced noise visible
Calibration *(optional)*	Temperature reads correctly at room temp
Analysis questions	Thoughtful answers demonstrating understanding